

Deep Tech Commercialization at Early TRL in Switzerland

Literature Review , Key Findings for the NCCR Muoniverse Innovation Platform

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Executive Summary

This document summarizes the key findings from a systematic literature review on deep tech commercialization at very early technology readiness levels (TRL 1–3), conducted for the NCCR Muoniverse Innovation Platform. The review covers 53 sources across two search rounds, including 17 core peer-reviewed papers, 6 institutional reports, and supporting material. The findings below are organized as actionable insights: what the evidence says, what NCCR Muoniverse can use, what to do, and what to avoid.

The headline finding: the space where NCCR Muoniverse operates, fundamental physics at TRL 1–3, is precisely where no existing commercialization framework was designed to work. Every framework assessed has a structural gap at this stage. The thesis contribution lies in identifying what support mechanisms fill this gap.

Headline Numbers

Metric	Value	Source
Sources reviewed	53	[1]–[53]
Deep tech share of Swiss VC	60–63% (highest globally)	[F5]
Switzerland spinout value per capita	#1 in Europe (index 100 vs runner-up 33)	[F5]
BRIDGE PoC founding rate	70% of funded researchers found a startup	[F1]
Innosuisse causal effect	+20.7% sales, +17.6% employment (5-year, DiD verified)	[F2]
Business advisor effect on readiness	+0.23 per advisor (p=0.04)	[N1]
Technology advisor effect on readiness	–0.31 per advisor (p=0.05)	[N1]
Market features as success predictor	$\beta = 0.514$ (strongest of all factors)	[T5]

1. Key Findings from the Literature

The 53 sources converge on seven evidence-based themes. Each finding below is stated with its supporting data and the source it derives from.

1.1 Deep tech is an ecosystem phenomenon, not a firm-level one

Raff-Heinen et al. (2026) analyzed 272 European regions and 1,605 deep tech startups. Two factors are statistically significant predictors of deep tech concentration: knowledge infrastructure (IRR = 1.48) and talent pools (IRR = 1.66). Formal institutional factors (regulation, government policy, legal frameworks) were NOT significant [T1]. The top 10 European regions account for 40% of all deep tech startups.

Seitz et al. (2026) meta-analyzed incubation programs: overall effect on venture performance $r = 0.067$, but university-affiliated incubators ($r = 0.118$), older incubators ($r = 0.183$), and technology-focused ventures ($r = 0.130$) benefit most. Networking activities show positive effects ($r = 0.071$); mentoring shows a negligible or slightly negative effect ($r = -0.012$) [T1].

1.2 TTOs must build relationships, not just process transactions

A 15-year longitudinal study of 231 Japanese university spin-offs found that TTO activities predicting venture survival, ordered by significance: close business relationships ($p = 0.0034$), customer introductions ($p = 0.006$), market knowledge transfer ($p = 0.015$), and technology knowledge transfer ($p = 0.042$). Relational and market-oriented support matters more than technical support [T2].

Chen et al. (2024) identified a 'playmaker' TTO typology, active orchestrators that span boundaries, build relationships, and de-risk commercialization, outperforming transactional 'go-between' TTOs. TTO specialization in specific technology domains produces a coefficient of 1.273 on spin-off performance vs. 0.035 for breadth (Martínez-Ardila et al., 2023) [T2].

1.3 Structured frameworks outperform ad hoc approaches

VentureWell (2019) identified six commercialization dimensions (team, technology, IP, market, business model, resources) across five stages, with an explicit 'abandon' pathway. Barron & Amorós (2020) showed that a 9-week accelerator requiring 100 customer interviews produces a paradigm shift in scientists' market understanding [T3].

Hayter et al. (2025) documented the NSF I-Corps model across 8 countries. The core methodology, lean startup + intensive customer discovery, delivered as a structured 7-week immersion, has been independently adopted in Germany (EXIST), Brazil, and six others. This constitutes a natural experiment confirming structured frameworks work, but every system adapted the model to local context [N3].

1.4 Support mechanisms must be configured, not just provided

Stahl et al. (2023) studied 14 deep tech startups and found three different effective configurations of five orchestration functions, with no single configuration universally optimal. The Seitz meta-analysis confirms: networking helps ($r = 0.071$) while more mentoring may not, and can even reduce performance through opportunity costs [T4].

The Hello Tomorrow survey (2019) found deep tech founders value mentors most (64%), followed by customer introductions (61%) and direct investment (37%). Capital ranks below market access and expertise, suggesting the European ecosystem's scarcest resources are not financial [T4].

1.5 Market identification is the binding constraint

This is the most practically important finding. Foresight (2024), in a regression of 110 European deep tech companies, found market features are the strongest predictor of commercialization success ($\beta = 0.514$), ahead of technology features ($\beta = 0.310$) and environmental factors ($\beta = 0.306$). Firm internal factors, team quality, management, were NOT significant [T5].

Cantner et al. (2024), studying 1,149 scientists with dominance analysis, found the commercial sphere that dominates the academic sphere in importance from the very outset of spin-off creation, not in later stages, from day one. This challenges the standard assumption that scientists should focus on research first and add commercialization later [N4].

Denoo et al. (2024) tracked 112 science-based ventures and found each additional business advisor increases 'application readiness' by 0.23 units ($p = 0.04$), while each additional technology advisor decreases it by 0.31 units ($p = 0.05$). Approximately four business advisors advance readiness by one full unit, equivalent to 15 years of additional entrepreneurial experience [N1].

1.6 Europe faces structural barriers

Stryber (2025) maps the gap: €58B European deep tech VC vs. €215B US (nearly 4×). 62% of European investors reject founders with prior failures vs. a US culture treating failure as learning. Only 15% of European PhD graduates pursue startups vs. 45% in the US. 12% of European corporations engage with deep tech startups vs. 40% in the US [T6].

Deep tech requires 35% more time and 48% more capital than conventional startups to reach equivalent milestones (UNDP, 2025). These interact with European structural deficits in compounding ways: the 35% additional time conflicts with typical VC fund cycles of 3–5 years [T6].

1.7 Convergence toward a deep tech definition

Kortsch et al. (2024) identified 12 defining attributes across three levels: technology (scientific origins, hardware intensity, platform character), venture (5–15+ year timelines, high capital intensity, multi-layered uncertainty), and ecosystem (knowledge dependence, talent concentration, infrastructure requirements). Cabanes et al. (2026) frame deep tech as a 'performative innovation regime' requiring patient capital, long-term partnerships, and ecosystem orchestration [T7].

2. What NCCR Muoniverse Can Retain from This Research

2.1 Evidence-based principles for support design

1. Prioritize business advisors over technology advisors. Evidence shows each business advisor increases application readiness by 0.23 units while each technology advisor decreases it by 0.31 units. Limiting technology advisors and prioritizing market/business advisors is counterintuitive but evidence-based for physics-based ventures. [N1]
2. Start commercial exposure from day one. The commercial sphere dominates the academic sphere in importance from the outset of spin-off creation, not after the technology is proven. The assumption that scientists should focus on research first and 'add commercialization later' is empirically wrong. [N4, T5]

3. Prioritize market identification as the binding constraint. Market features ($\beta = 0.514$) are the strongest predictor of commercialization success, ahead of technology quality ($\beta = 0.310$). For muon technology, the critical success factor is identifying market applications, not the scientific excellence (which is world-class by definition) or funding availability (which NCCR status provides). [T5]
4. Build relational TTO support. Close business relationships maintained by TTOs predict venture survival more strongly ($p = 0.0034$) than any other TTO activity, including technology knowledge transfer. PSI's TTO should adopt a 'playmaker' orientation (active orchestrator) rather than a 'go-between' (transactional intermediary). [T2, N2]
5. Leverage university-affiliated, specialized incubation. University-affiliated incubators ($r = 0.118$) and older incubators ($r = 0.183$) substantially outperform generic ones. PSI's domain specialization in physics is an asset that should be deepened, not diluted. [T1, T2]
6. Use structured frameworks with explicit abandon/pivot pathways. The sunk cost fallacy is particularly dangerous when development timelines stretch over 5–15 years. Frameworks should include explicit 'abandon' and 'pivot' decision points, not just 'advance'. [T3]

2.2 Swiss ecosystem assets NCCR Muoniverse can leverage

- Switzerland ranks #1 in Europe for spinout value per capita (index 100 vs runner-up's 33) and #3 in absolute enterprise value (\$44.6B). [F5]
- ETH Zurich ranks #3 in Europe for spinout value; EPFL #4; UZH #8. Three Swiss institutions in the top ten. [F5]
- 60–63% of all Swiss VC flows to deep tech, the highest proportion globally, ahead of China (56%) and the US (54%). [F5]
- ETH introduced 2% equity stakes and a 6–8 week spin-off IP licensing process, among the most founder-friendly terms in Europe. [F-Eth]
- EPFL Innogrant (CHF 100,000 for one year of salary) complements Innosuisse Initial Coaching and can precede it. [F-EPFL]
- Swiss startups raised CHF 1.47B in H1 2025 alone (+36% YoY); total Swiss VC reached CHF 2.95B in 2024. [F5]

2.3 CERN as the closest institutional analog

CERN faces the identical commercialization challenge as NCCR Muoniverse: fundamental particle physics → commercial applications. The CERN Venture Connect (CVC) model takes zero equity, offers founder-friendly express agreements, curates technology from CERN's portfolio, and connects startups to 60 partner organizations across 18 countries. Eight startups joined CVC in 2025, collectively raising over €5M [F3]. This is the most directly comparable institutional model and should be benchmarked in Part 2.

2.4 PSI spin-off ecosystem

PSI hosts one of Switzerland's most productive physics-based spin-off ecosystems, with 20+ spin-offs. DECTRIS AG (2006) commercialized PSI's PILATUS X-ray detector technology and is now a global leader, the exact trajectory (particle physics detector → commercial product → global market) NCCR Muoniverse hopes to enable for muon detection. Other spin-offs span neutron optics (SwissNeutronics, 1999), nanolithography (EULITHA, 2006), metal-organic frameworks (novoMOF, 2017), and grating interferometry (GratXray, 2017) [F4].

2.5 Innosuisse evidence for what works

Instrument	What works (evidence)	Source
BRIDGE PoC	70% founding rate , highest of any Innosuisse instrument. CHF 130K over 12 months.	[F1]
Innovation Projects	57% market launch within 3 years. 4× leverage (1 CHF → 4 CHF value). 2.5 new jobs/project.	[F1]
Core Coaching	78% satisfaction. 21% raised >CHF 5M. 55% radical innovation rate.	[F1]
Causal effect (KOF)	+20.7% sales, +17.6% employment over 5 years. Strongest for small firms (5–50 employees) with ETH Domain partners.	[F2]

3. Gaps in the Literature the Thesis Will Address

The review identified gaps that represent the thesis's opportunity for original contribution. These will be addressed through firm interviews (Part 3) and cross-method synthesis (Part 4).

Gap	Status after Round 2	Thesis contribution
Swiss-specific deep tech research	Partially filled (KOF, DTN, Innosuisse)	Peer-reviewed Swiss analysis remains thin; interviews add primary evidence
Early-TRL (1–3) processes	Partially filled (Hayter, Denoo)	Interviews probe what specifically happens at earliest stages
Longitudinal tracking	Partially filled (15-year TTO panel, Cantner)	Interviews add retrospective trajectories
Failure analysis	Still absent	Critical gap , interviews will probe abandonment decisions
Sector-specific (physics)	Filled (CERN, PSI, DECTRIS)	Interviews add muon-specific evidence
Ecosystem interaction effects	Partially filled	Cross-method synthesis maps mechanism combinations
Measurement & metrics	Partially filled (application readiness)	Benchmarking proposes assessment criteria
Policy design principles	Partially filled (16-mechanism playbook)	Part 4 synthesizes design recommendations

4. Next Steps

- Interview protocol (Part 3, milestone September 1, 2026): probe the gaps above, particularly the Swiss-specific gap, failure analysis, and physics-specific commercialization. John Millard (Head of TTO, PSI) is the primary interview target.

Sources

All sources were directly accessed and verified during June–July 2026. No source has been fabricated or inferred.

Core Academic Papers

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- [T2] Kruachottikul et al. (2023). 231 Japanese university spin-offs, 15-year panel. TTO relationships $p=0.0034$. *Journal of Technology Transfer*.
- [T2] Chen, Z., Little, V. & Thuan, N. (2024). Go-betweens or playmakers? *J. Technology Transfer*, 50, 1060–1079.
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- [T3] VentureWell (2019). Six dimensions, five stages, explicit abandon pathway. 500+ practitioners.
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- [T4] Stahl et al. (2023). 14 deep tech startups, 5 orchestration functions, 3 effective configurations.
- [T4] Hello Tomorrow (2019). Global deep tech survey. Mentors 64%, customer intros 61%, investment 37%.
- [T5] Foresight (2024). 110 European deep tech companies. Market $\beta=0.514$, technology $\beta=0.310$. *J. Innovation & Knowledge*.
- [T6] Stryber (2025). European deep tech barriers. €58B vs €215B VC. 62% reject failed founders.

- [T6] UNDP (2025). Deep tech: 35% more time, 48% more capital than conventional startups.
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- [N3] Hayter, C.S. et al. (2025). Launching lean: I-Corps across 8 countries. J. Technology Transfer.
- [N4] Cantner, U. et al. (2024). Small Business Economics, 62, 1555–1590. 1,149 scientists. Commercial sphere dominates from day one.

Institutional and Policy Sources

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- [F2] Hulfeld, F., Spescha, A. & Wörter, M. (KOF ETH Zurich) (2024). Funding R&D Cooperation , Innosuisse effectiveness. DiD: +20.7% sales, +17.6% employment.
- [F2] KOF/ZEW (2024). Monitoring des Wissens- und Technologietransfers in der Schweiz. KOF-Studie Nr. 179, commissioned by SBFI.
- [F3] CERN (2025). Knowledge Transfer Highlights 2025. 89 KT contracts. CVC: 0% equity, 60 partners, 18 countries.
- [F4] PSI. Spin-off ecosystem. DECTRIS AG, SwissNeutronics, EULITHA, novoMOF, GratXray. John Millard, Head of TTO.
- [F5] Deep Tech Nation Switzerland / Dealroom.co (2025). European Spinout Report. Switzerland #1 per capita, ETH #3, EPFL #4.

[F-Eth] ETH Board. Common spin-off licensing principles. 2% equity, 6–8 week process.

[F-EPFL] EPFL Innogrant. CHF 100,000 for one year of salary.

[FA-EIC] European Innovation Council. Switzerland fully associated since January 1, 2025.

[FA-CRL] Abbas, A. & Nomvar, M. Commercial Readiness Level (CRL) Framework. University of Sydney / Scimita Ventures.